ECON 0150 | Final Project

Guidelines

The final project is your place to demonstrate your ability to analyze and communicate data, using at least one skill / concept from each of the six parts. You will select a dataset, conduct a comprehensive analysis, and present your findings through both a slide presentation and a written report. I'll select some projects to be displayed on the course website after the semester has finished and used as references for future classes.

The most challenging steps in the final project are typically 1) finding a question that's answerable with data, and 2) finding a dataset on the vast internet. I've provided some example projects from past semesters and a list of good economic data sources (e.g., FRED, World Bank, BLS, Election Data Source, etc.). You're more than welcome to 1) use data from one of the provided replications or databases, or 2) find or collect your own dataset related to an economic question of interest to you. If you choose to use a dataset from a previous semester's group, you need to demonstrate that you, and not just the previous group, can analyze and communicate data.

The project will build progressively over the semester. At the end of each Part, you will submit a part project demonstrating your ability to apply the concepts in that Part to a dataset of your choosing. You are encouraged to work in a group and to pick a topic that could turn into your final project. But you are more than welcome to switch groups or topics at any point.

Make this your own and have fun with it! There are general guidelines, but don't let the guidelines stop you from doing what interests you. Lets just have a quick chat about it.

Timeline

Parts 1 & 2 Project | *Due Friday, October 16 at 5PM*

This semester the Part 1 and Part 2 projects are combined into a single submission. Using a dataset of your choosing, include:

1. Your group members: 1-2 others (groups of 2-3). Submit one group submission on Gradescope with everyone included.
2. A tentative question you'd like to answer with your data, and where your data comes from.
3. A diagram of your dataset: what each row represents, the main columns, and each variable's type (*categorical or numerical; cross-sectional or time series*).
4. A summary of a single variable using an appropriate Part 1 tool (*bar chart, histogram, boxplot, line plot, etc.*), with a sentence or two describing what it shows.
5. A visualization of a relationship between two or more variables using an appropriate Part 2 tool (*scatterplot, multiboxplot, grouped bar chart, etc.*), with a sentence or two describing what it shows.

Part 3 Project

- There is no Part 3 project.

Part 4 Project | *Due date TBA*

- Build a regression model using the concepts in Part 4. Include:
  1. Your group members: 1-2 others (groups of 2-3)
  2. The equation for your model.
  3. A visualization of the core relationship in your model (*scatterplot, multiboxplot, etc*)
  4. A fitted model and its summary.
  5. An interpretation in words of the β_0 and β_1 coefficients and their p-values. Be as specific as you can. Write this so that someone who doesn't understand your data can understand the model results.

Part 5 Project | *Due date TBA*

- Build a regression model using a control variable. This can build on your Part 4 Project. Include all the same components as the Part 4 Project but with your updated model.

Later

- *Draft analysis (date TBA): Send me a draft of your writeup and slides, including at least one figure.*
- *Final presentation (during the final exam period, date TBA): Present 1-2 slides, described below.*
- *Written report and replication folder (due at the end of finals week, date TBA), described below.*

Written Report (1-2 pages)
Submit a short report (roughly 2 pages) and a folder with all your data and replication code. Your report will include the following:

1. Introduction and question

   Successful introductions quickly introduce a question that is both answerable with data and is an economics question. Focus on something that you are interested in and keep the question small enough that it is something you can answer. You do not need to ask a world-changing question at this stage in your career.

2. Data description and sources

   Include summary statistics and data visualizations of your main variables and relationships. Mention where your data comes from, any issues with reliability and source credibility. Describe any data cleaning or preparation needed for your analysis.

3. Methodology

   Describe your regression model, what each part of the model is doing, and its limitations. This should not be long, potentially as short as a single paragraph.

4. Results and analysis

   Report a summary of your model's results. Interpret the important parts of your results in words. Assess the validity of the model using a residual plot. Mention any limitations of your model.

5. Conclusions

Like your introduction, tell your readers what you did, what you found, and how it answers your question. Make it easy for your reader to understand the connection between your analysis and your question.

6. References

   Include references to your data sources and any relevant literature. There is no length or format requirement. The references section is a place to allow those reading your work to be able to replicate it.

Presentation (1-2 slides)

Summarize your research in 1-2 clear, concise slides highlighting:

- Research question
- Key findings
- Most compelling visualization
- Main conclusions

Be prepared for a 3-minute presentation during the final exam period.

Replication Folder

Submit a link to a folder with everything someone else would need to replicate your findings. The main purpose of the folder is to make it clear where your data comes from and how you produced your figures and model results. Your folder should include:

1. Your data

   Include the raw data files you downloaded or collected, along with any cleaned versions you created. If a dataset is too large to include, link to where it can be downloaded.

2. Your notebook

   Include the notebook you used to clean your data, make your figures, and run your model. Someone should be able to open it and run it from top to bottom to get your results. The Project Template Notebook on the course website is a good starting point.

3. A short README

   A few sentences describing what's in the folder, where each dataset comes from (with links), and the order to run things in.

4. Other materials (optional)

   Figures, your slides, and your report are welcome here too.

The format isn't strict. Most groups share a Google Drive folder. If you do, make sure I can open it: right click on your folder, click "Share", and switch "General access" from "Restricted" to "Anyone with the link". Submit one link per group on Gradescope with all group members included. If you'd prefer that I not use your name if your project is shared as an example for future classes, just let me know.